

DIGITAL TWINS — GROUP ASSIGNMENT 2

From Reactive Fire Detection to Predictive and Coordinated
Emergency Intelligence

FIREGUARD DT

Intelligent Multi-Twin Ecosystem for Fire Prediction,
Evacuation & Emergency Response

ONE LIVING DIGITAL MODEL COORDINATES PREDICTION, EVACUATION, AND EMERGENCY RESPONSE.

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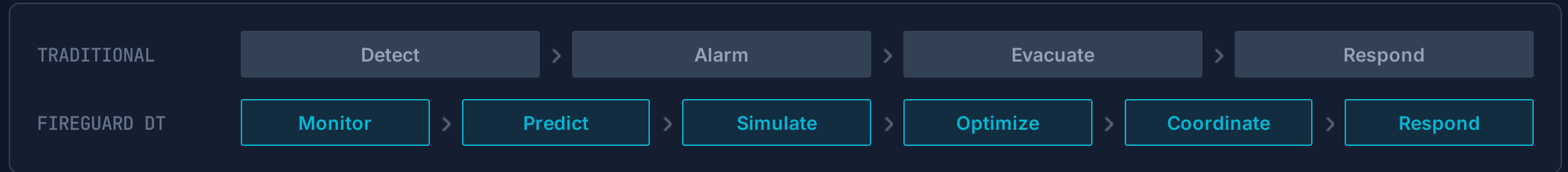
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THE PROBLEM: TRADITIONAL FIRE RESPONSE IS REACTIVE

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Fire emergencies evolve faster than static safety plans.



Late Awareness

- Alarms react **after** abnormal conditions are detected.
- Teams receive limited forward-looking insight into risk.

Static Response

- Predefined evacuation routes are rigid and non-adaptive.
- Failure to account for smoke, congestion, or blocked exits.

Fragmented Intelligence

- Building, occupancy, and environmental data remain isolated.
- Lack of unified cross-domain situational awareness.

⚠ Trust Challenge: Unexplained AI predictions create governance and accountability concerns.

Can a Digital Twin predict emerging risk and continuously coordinate the safest response before conditions become critical?

FIREGUARD DT — INTELLIGENT MULTI-TWIN ECOSYSTEM

4 INTERACTING DIGITAL TWINS + AI DECISION ORCHESTRATOR + SHARED 3D SPATIAL MODEL

Fire & Environment Twin

- Predicts emerging fire risk profiles
- Tracks Temp, Smoke, CO/CO₂, Humidity
- Monitors electrical load & HVAC effects

Occupancy & Evacuation Twin

- Real-time people distribution mapping
- Congestion & vulnerability assessment
- Calculates dynamic safe evacuation routes

Building Infrastructure Twin

- Rooms, corridors, and exit status
- HVAC zones & sprinkler health
- Power grid & critical asset monitoring

Emergency Response Twin

- Coordinates fire crews & drones
- Identifies optimal access points
- Resource allocation & dispatch management

CENTRAL INTELLIGENCE HUB AI / DECISION ORCHESTRATOR

 State Fusion

Prediction

Safety Rules

 Optimization

 Explainability

 Oversight

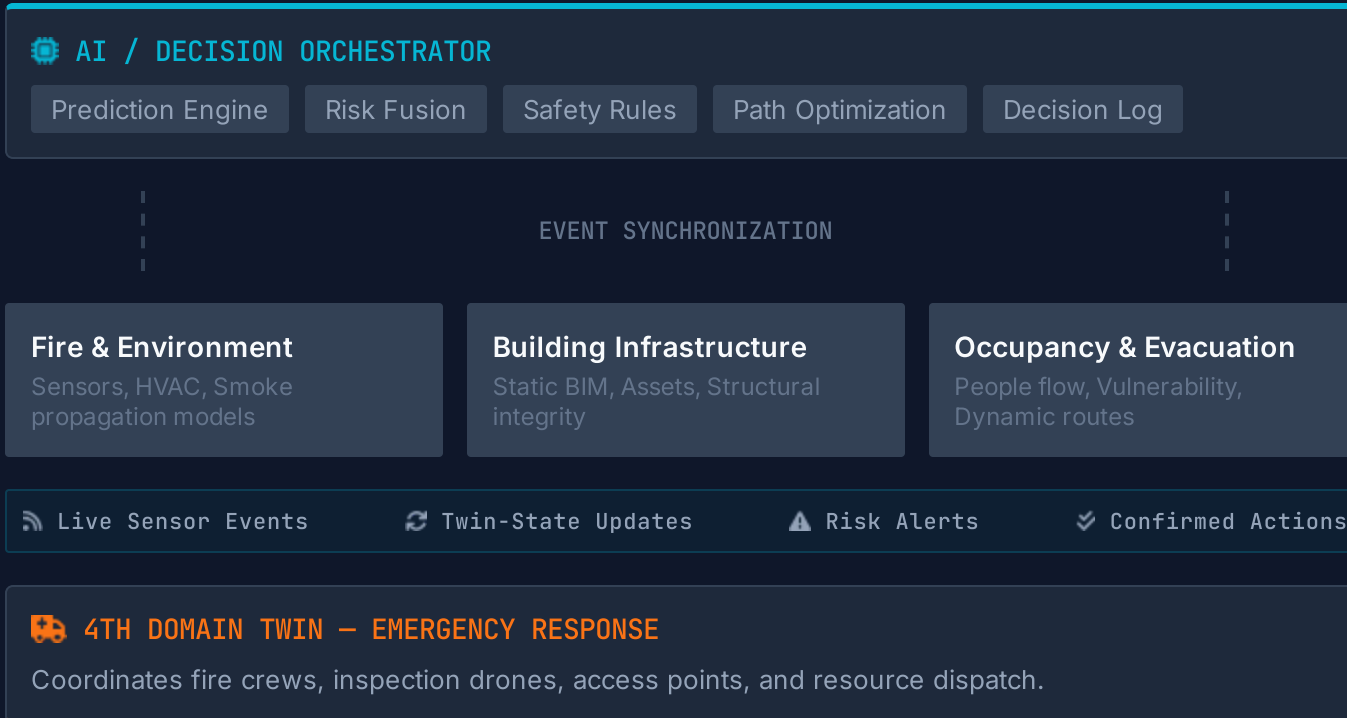
Cross-twin decision layer

Fuses state, predictions, safety rules, optimization, and human oversight across the four domain twins.

4 domain twins + AI orchestration + shared 3D spatial context — not a fifth twin > one coordinated response

INTEGRATED MULTI-TWIN ECOSYSTEM

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Coordination Strategy

HYBRID / FEDERATED ARCHITECTURE



Domain Autonomy

Each twin maintains its domain-specific state and specialized simulation models independently.



Event + Spatial Backbone

Twins exchange events through a shared backbone and interpret them against one common 3D building context.



Central Orchestration

A centralized AI layer fuses multi-twin states to coordinate complex, cross-domain decisions.



Safety Governance

Critical actions are strictly filtered through deterministic safety rules and human approval gates.



Assignment Evidence: four logical twins + mapped event flow + shared 3D spatial context + governed coordination.

3D EMERGENCY SCENARIO: FROM PREDICTION TO ACTION

SHARED 3D SPATIAL MODEL CONNECTS HAZARDS, INFRASTRUCTURE, PEOPLE, ROUTES, AND RESPONSE RESOURCES

01

⚡ PREDICT

Electrical-room sensors:
Temp ↑, Smoke ↑, CO ↑

87%

CRITICAL RISK

**Illustrative value*

02

📦 SPATIALIZE

Shared 3D Model maps:
Elec. Room → Corridor C
→ HVAC Zone 3

Places the hazard and
smoke-propagation path
inside one common
building context.

03

👤 LOCATE PEOPLE

Occupancy Twin locates
people:
within affected 3D zones

43

Occupants at Risk

**Illustrative value*

04

🧭 OPTIMIZE

Smoke makes Corridor C
unsafe.

Path A: Rejected

Path B: Selected

Dynamic recalculation
based on current hazard
state.

05

🚒 RESPOND

Response Twin:
Dispatch drone & crew

2m 31s

Illustrative ETA

**Illustrative value*

06

✅ COORDINATE

Orchestrator:
Validate safety rules
Request human approval
& log decision trail.



Audit Log Updated

📦 SHARED 3D BUILDING MODEL

COMMON CONTEXT — NOT A FIFTH
TWIN



Electrical Room



Smoke / Corridor C



43 occupants



Safe Route B



Crew access

PREDICTIVE INTELLIGENCE

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01 // DATA INPUTS

- Temperature
- Smoke Conc.
- CO / CO₂
- Humidity
- Electrical Load
- Occupancy
- HVAC State
- Sequence Data

02 // MODEL PIPELINE

- Logistic Regression (Baseline)
- Random Forest Classifier
- Gradient Boosting / XGBoost

EVALUATION METRICS

Accuracy	TBD
Precision	TBD
Recall	TBD
F1-Score	TBD
ROC-AUC	TBD

03 // ACTIONABLE OUTPUT

NORMAL	8%
WARNING	58%
CRITICAL	87%

Est. Time-to-Critical: 6.3 min

Evaluation Principle:

Prioritize **Recall** for dangerous conditions. A missed hazard is more costly than a false warning.

 **Evidence to Insert:** Final submission will replace TBDs with confusion matrix, model comparison charts, and test results from the AutoML notebook.

EXPLAINABLE & TRUSTWORTHY AI

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⚠️ CRITICAL ALERT

87% Risk

Illustrative scenario value — not a measured result

EXPLAINABILITY METHODS

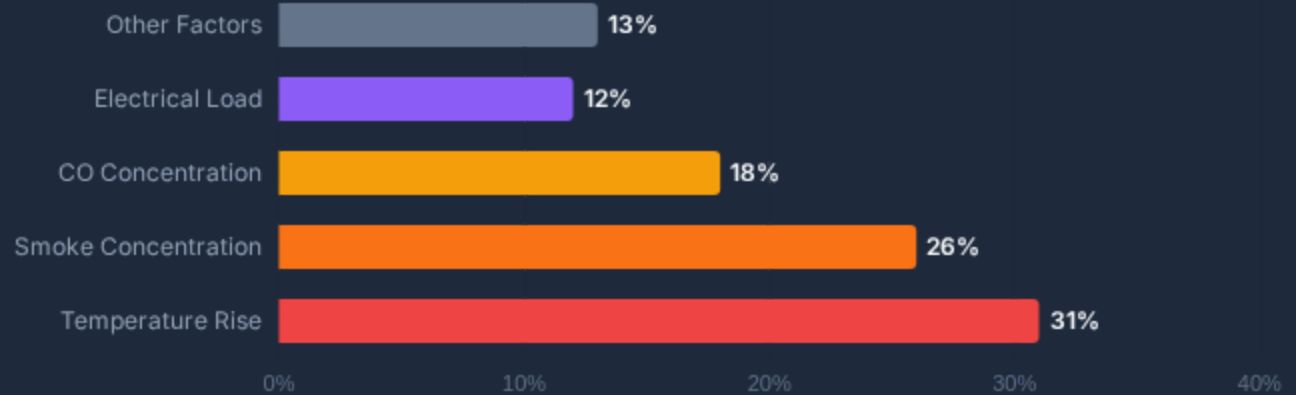
- Global feature importance
- Local SHAP explanation
- Model confidence & threshold
- Physical-consistency check against sensor patterns

CONTROLLED-AUTONOMY LADDER



Risk Contributor Breakdown — Fire Risk 87%

🧪 Illustrative Values



OPERATOR QUESTIONS ANSWERED

- ❓ Why was this prediction generated?
- 📊 Which signals contributed most?
- 🔗 Is the prediction consistent with physical conditions?

DYNAMIC EVACUATION OPTIMIZATION

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TRADITIONAL APPROACH

Static paths optimized for distance. Fails when routes are blocked by smoke or fire.

FIREGUARD DT

Dynamic paths optimized for safety. Adapts to real-time environmental hazards.

$$C(P) = w_1 D(P) + w_2 R_{\text{fire}}(P) + w_3 R_{\text{smoke}}(P) + w_4 C_{\text{crowd}}(P)$$

$D(P)$: Distance

$R_{\text{fire}}(P)$: Fire Risk

$R_{\text{smoke}}(P)$: Smoke Risk

$C_{\text{crowd}}(P)$: Congestion

$$P^* = \arg \min_p C(P)$$

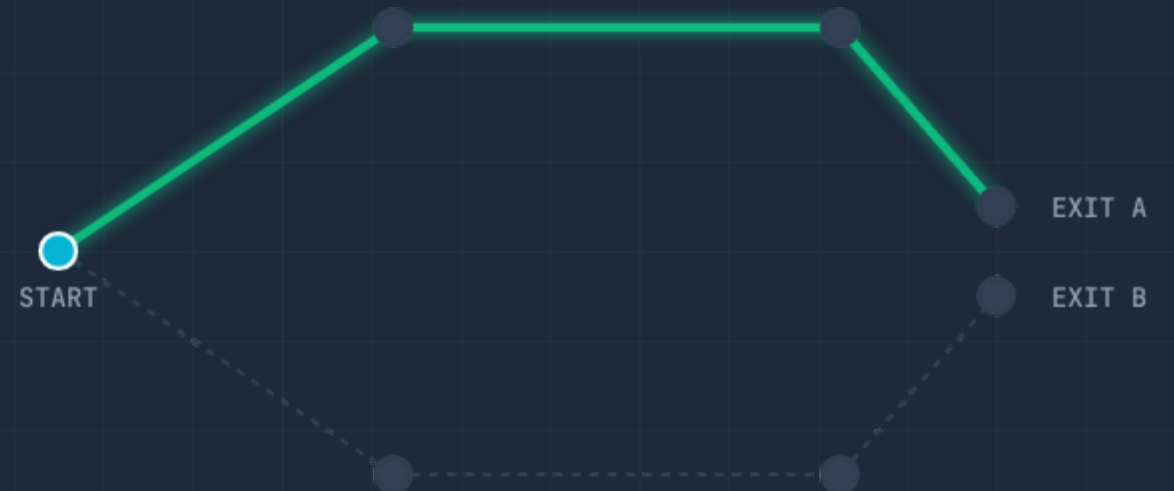
- Algorithm: Dynamic Dijkstra / A*
- Constraints: Blocked zones = ∞ cost
- Priority: Vulnerable occupant safety

EXECUTIVE TAKEAWAY

Shortest routes are often the most dangerous. FireGuard DT ensures the safest path is always prioritized.

DYNAMIC ROUTE RECALCULATION

REPLAY SIMULATION



ROUTE: OPTIMAL

3D WHAT-IF SIMULATION LAB

ONE SHARED 3D BUILDING CONTEXT FOR HAZARDS, SMOKE, PEOPLE, ROUTES, AND RESPONSE RESOURCES

3D Scenario Configuration

COMMON SPATIAL CONTEXT — NOT A FIFTH TWIN

FIRE ORIGIN

 **Electrical Room**

FIRE SEVERITY

 **High**

OCCUPANCY

 **250 Persons**

EXIT B STATUS

 **Blocked**

HVAC STATE

 **Running**

SPRINKLER

 **Failed**

Scenario Library

 Normal Response

Blocked Exit

 Peak Occupancy

 Smoke Propagation

 Sprinkler Failure

 Sensor Malfunction

STRATEGY	EVACUATION TIME	RISK EXPOSURE	PEAK CONGESTION
Static Plan	TBD	TBD	TBD
Shortest Path	TBD	TBD	TBD
Twin Optimized	TBD	TBD	TBD

Spatial Simulation Outputs

 Render fire origin, smoke/HVAC propagation, blocked areas, and infrastructure state in the shared building model.

 Recalculate people movement, safe routes, congestion, and crew access against the same spatial conditions.

 Compare evacuation time, dangerous-zone exposure, peak congestion, and response time across strategies.

Key Message: The shared 3D spatial model lets the team test coordinated safety strategies before applying them to the physical environment.

CYBERSECURITY & RESILIENCE

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Device & Communication

- › Device identity & authentication
- › TLS-encrypted communication
- › Signed / secured APIs



Platform Security

- › Role-Based Access Control
- › Network segmentation
- › Audit logging
- › Encryption at rest



Sensor Trust

- › Multi-sensor consistency checks
- › Plausibility / range checks
- › Anomaly detection
- › Source & timestamp validation



Resilience

- › Edge safety rules
- › Local alarm continuity
- › Graceful degradation
- › Recovery & sync after reconnection

⚠️ ILLUSTRATIVE ANOMALY DETECTION SEQUENCE

Sensor A — Temperature: 95°C ↑

Sensor B — Temperature: 26°C ✓

Smoke: Normal · CO: Normal

→ Potential sensor anomaly detected

→ Request validation — do not auto-declare fire from one inconsistent sensor

🔒 THREAT-TO-CONTROL MAPPING

🚩 Spoofed sensor → Multi-sensor consistency checks

🏠 Compromised account → RBAC + MFA enforcement

📶 Network outage → Edge fallback & local continuity

📜 Tampered decision → Signed immutable audit trail

GOVERNANCE, ETHICS & RESPONSIBLE AUTONOMY

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Transparency

- Explainable predictions (SHAP)
- Confidence level reporting
- Immutable decision audit trails



Privacy

- Anonymous occupancy counting
- PII-free data minimization
- Secure zone-based isolation



AI Safety

- Mandatory human-in-the-loop
- Deterministic rule validation
- Fail-safe edge operation



Bias & Reliability

- Zone-specific model evaluation
- Vulnerable occupant priority
- Cross-scenario reliability tests

RISK-BASED AUTONOMY MATRIX

DECISION TYPE	RISK LEVEL	RECOMMENDED CONTROL MECHANISM
System Health & Dashboard Updates	LOW	Fully Autonomous background processing
Evacuation Route Recalculation	MEDIUM	Autonomous with deterministic rule-safety validation
HVAC / Smoke-Control Mode Change	HIGH	Strong validation + Mandatory authorized approval
Critical Resource Dispatch (Crews/Drones)	HIGH	Human-in-the-loop + Multi-stakeholder audit trail

Governance Principle: Automation level is strictly proportional to decision risk. AI supports emergency decisions; it does not replace human responsibility.

BUSINESS VALUE AND ROI

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Safety Benefits

- Earlier risk identification through predictive analytics
- Faster emergency response with coordinated twins
- Safer evacuation via dynamic route optimization

Operational Efficiency

- Reduced false alarms and unnecessary evacuations
- Predictive maintenance for critical safety assets
- Accelerated incident investigation and reporting

Financial Protection

- Reduced business disruption and downtime
- Potential property damage and insurance reduction
- Optimized resource utilization during emergencies

ROI Methodology

$$ROI = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}} \times 100\%$$

INVESTMENT COSTS

IoT Sensor Integration
Twin Platform Licensing
ML Model Development
Security Hardening
Staff Training & Drills

PROJECTED BENEFITS

Avoided Downtime
Damage Risk Reduction
Maintenance Savings
Response Efficiency
Risk Compliance Value

Scenario Analysis

Metric	Conservative	Base	Optimistic
Initial Investment	TBD	TBD	TBD
Annual Benefit	TBD	TBD	TBD
Payback Period	TBD	TBD	TBD
3-Year ROI	TBD	TBD	TBD

Note: Values are project assumptions for feasibility analysis and must be clearly distinguished from observed data.

STRATEGIC DEPLOYMENT ROADMAP

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PHASE 01

Digital Pilot

-  1 Floor deployment
-  Fire + Building Twins
-  Simulated sensor nodes
-  Initial ML baseline

PHASE 02

Intelligent Pilot

-  Full Building scope
-  Occupancy Twin integration
-  Dynamic evacuation engine
-  Explainable AI module

PHASE 03

Operational

-  Response Twin activation
-  Full IoT sensor network
-  Edge intelligence nodes
-  Security hardening

PHASE 04

Campus Ecosystem

-  Multi-building expansion
-  Shared response resources
-  Central Command Center
-  Cross-building sync

PHASE 05

Intelligent Ecosystem

-  Continuous ML monitoring
-  Predictive maintenance
-  Resource optimization
-  Controlled autonomy

VALIDATION STAGE GATES

Model

Target recall/F1 and false-alarm tolerance

Safety

Successful scenario tests & rule verification

Security

Pen-testing & incident recovery exercises

Operations

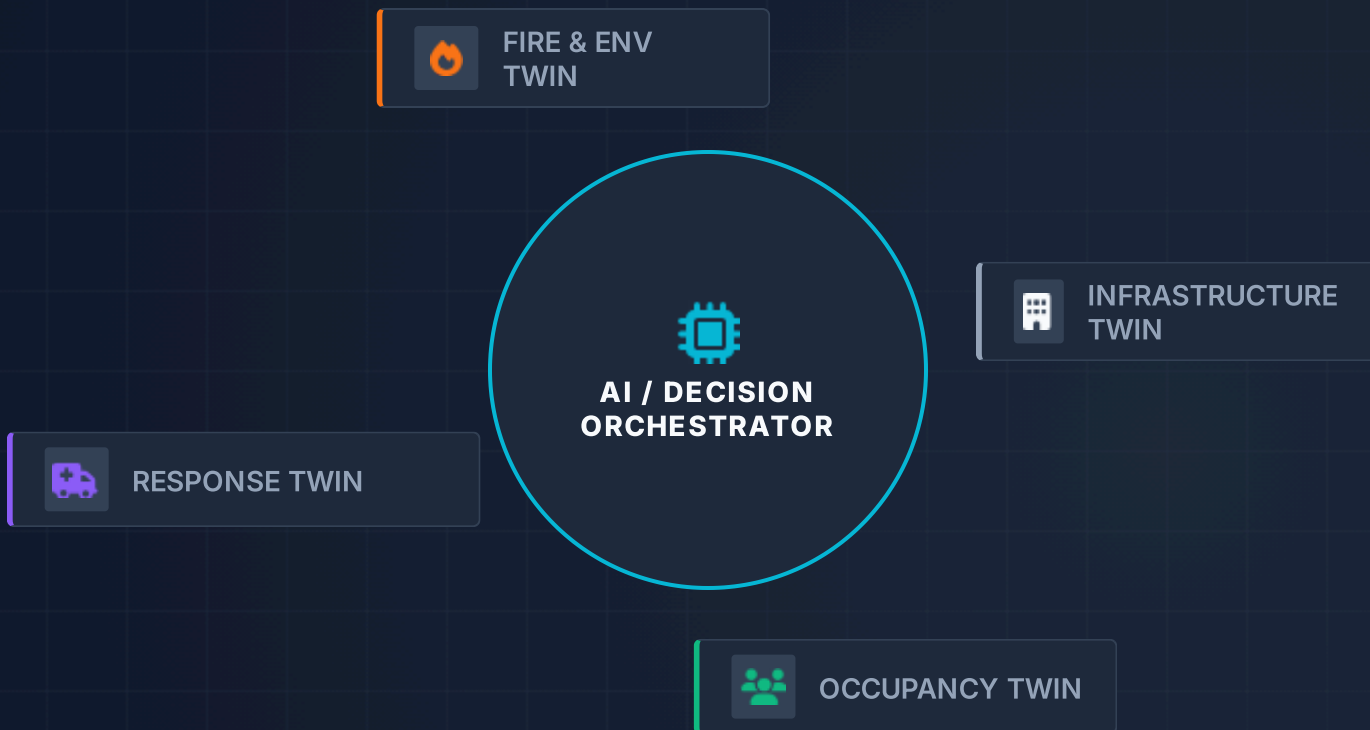
Staff training & emergency-drill acceptance

Business

Approved ROI assumptions & pilot evidence

RESULTS AND CONCLUSION

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⚡ PREDICT

Identify emerging fire risks early.

👁️ UNDERSTAND

Unified view of building context.

🔧 OPTIMIZE

Safest dynamic evacuation paths.

🔄 COORDINATE

Sync resources and response.

🛡️ GOVERN

Trust through explainability.

FireGuard DT transforms a fire alarm from a reactive warning into a predictive, coordinated, and continuously adapting emergency-response ecosystem.

PERFORMANCE METRICS

System Impact

PREDICTION ACCURACY (F1)

TBD

PENDING

EVACUATION IMPROVEMENT

TBD

PENDING

RESPONSE TIME REDUCTION

TBD

PENDING

PROJECTED 3-YEAR ROI

TBD

PENDING

Note: Final submission must replace placeholders with reproducible outputs from the model notebook, scenario simulations, and documented ROI assumptions.